

The Hand-Driven Approach

How a gold-standard Casually Explained window is built by hand,
and exactly how failures get diagnosed and fixed.



casually_engine_v7 | chapter-1 pipeline | written from the working repo, not from memory

1. What hand-driven means

A hand-driven window (W01-W03, the 30-second demos that set the bar) is 4-5 hand-written skit functions, e.g. w1a..w1e in scripts/render-gag-w01.ts. Each skit is a GagDef: a start time, an end time, and a function `fn(t, lt)` that returns the complete frame at that instant: background SVG, cast states, foreground props, and camera.

Nothing is keyframed and nothing is generated. For every frame at 24 fps, code computes where everything is. Same code always yields the same pixels, so a fix is verifiable by re-render, not by luck.

The shared grammar lives in scripts/gag-lib.ts (stage, props, walkers, camera, the driveGags renderer). Motion math lives in src/motion/ (eases, squash). Characters live in src/character/ (bodies, hairstyles, outfits, expressions). A window script only stages skits from these parts.

The five layers one author controls

- SVG assets: prop builder functions returning SVG strings (moneyBag, terminal, flashBurst, ...). House rule: zero words anywhere in frame.
- Character animation: per-frame StickFigureState for every actor: position, talking, walking, blink, expression, pose, scale, rotation, lean, gaze and pointing targets, hand props.
- Camera: per-frame GagCam {cx, cy, zoom, cut}. Wide 1.25 for setup, punch 1.5 / macro 1.6 for payoff. A Camera object smooth-follows targets; cut=true snaps.
- Cuts: beat boundaries are hard cuts (cut on the first frames). Punch moment cuts wide to tight. No dissolves, per house style.
- Audio: the voiceover MP3 is sliced per window (-ss start, -t duration) and muxed with -shortest as AAC. Cue times come from the SRT; audio never drives animation except through cue timing.

2. The time model (the core discipline)

Every skit function receives `t` (absolute seconds) and `lt` (seconds since its own start). All motion is a pure function of one of these via small helpers. There is no hidden state between frames, which is what makes misbehaviour diagnosable: any frozen frame can be reproduced by calling `fn(t, lt)` once.

```
seg(lt, a, b)      = clamp((lt-a)/(b-a), 0, 1)    # progress of a cue window
easeIO(k)          = k*k*(3-2*k)                 # smoothstep in/out
lerp(a, b, k)      = a + (b-a)*k                 # travel between marks
walkerX(lt,x1,a,b,c,d) = enter -200->x1 over [a,b], exit x1->2150 over [c,d]
walkerXR(lt,x1,a,b)  = enter 2100->x1 over [a,b], then stays
blinkAt(t)          = (t % 3.7) < 0.12           # 120 ms blink
popIn(k)            = overshooting scale-in for props (src/motion/eases)
```

The house beat shape is setup then payoff: wide shot while the host talks and business happens (walk-ins, prop cues at fixed `lt` times), a punchAt at about 70% of the beat, then a cut to tight with a gesture (point, lean, scale pop) and a guest reaction. W01 skit A does exactly this: host talks until `lt` 6.5 of 8 s, phone slab drops at `lt` 2.0-2.6, hearts rise at 3.1/3.5/3.9, punch at 6.5 flips expression to smug_thumbs_up with a point, camera cuts to tight.

Talking, entrances, expression flips

- Talking: the host carries the voiceover, so the host mouth moves nearly the whole beat (W01: isTalking: `lt` < 6.5 of 8 s). Guests talk only briefly, as reactions on the payoff.

- Entrances: a walker with a real duration (e.g. enter over [0.4, 1.0]) reads as arrival; a zero-duration entrance reads as teleportation or, worse, a figure that was just always there. Everyone who matters should visibly arrive.
- Expression flips: holding one expression per beat reads as frozen. The punch must change the face (deadpan -> smug/shock) AND the body (point, lean, pop) at the same instant.
- Idle life: while standing, figures sway (rotation sine ~ 4 deg) and breathe (scale sine $\sim 1.5\%$). Below conscious notice, but its absence is instantly felt as 'still image'.

3. How mess-ups get solved (failure catalog)

Every entry below is a real incident from this pipeline: symptom, root cause as found in code, and the exact fix. This is the debugging method in full: reproduce against the real code, trace to the frame function, fix the root cause across every case it implies, then add the gate that would have caught it.

3.1 Statue figures (the W04 incident)

Symptom: a 31-second window of a stick figure standing still. The staging critic passed it anyway. Root causes, all in the plan path at once: the host entrance had zero duration (start == end) so walkerX returned 'on, never walking'; only the host talked and only pre-punch; expressions were frozen per beat; props popped in and sat static; the camera was two static holds with one hard cut. Fix: real 1-second walk-in entrances; host talks the whole beat except a breather after the punch; guests react-talk on the payoff; idle sway plus breathing on every figure; scale pop and spine lean at the punch; a slow push-in under every setup. Gate added: audit-motion (section 4).

3.2 Empty tableau

Symptom: a beat with one lonely actor, no prop, no effect. Reads as dead air. Fix: the style gate now fails any beat with a lone actor and zero props and zero cast fx; propless tableau beats give the host an intent effect (question marks) so the frame always says something.

3.3 Dead air inside windows

Symptom: beats not covering their window; the concat then holds a frozen tail. Fix: timing gates allow up to 2.5 s of leading silence (real) but fail true gaps inside a window; splitBeats merges trailing fragments under 4 s into the previous beat.

3.4 Machine-gun cutting

Symptom: adjacent beats all punching, no breathing room. Fix: the compiler alternation pass demotes every second adjacent punch to wide (staging kept, only the shot changes), and the shot-budget backstop caps non-wide shots per window.

3.5 Host outside the tight frame

Symptom: punch-in crops the speaker out. Fix: the coverage gate computes the visible frame width from the zoom (1920/zoom) and fails any non-wide beat whose host mark lies outside it; tight cx defaults to the cast centroid, clamped to 700-1300.

3.6 Actors sharing a mark

Symptom: two figures drawn on top of each other. Fix: the collision gate fails any pair closer than 220 px; guests take fixed marks 680/990/1300/1620 with nearest-edge entrances so nobody crosses the cast.

3.7 Silent self-sabotage in code

Three incidents where the code looked right and the output was wrong. (a) Duplicate object key: a later 'scale: c.scale ?? 1' silently overwrote the breathing scale computed two lines above, so figures held exactly 1.0000. Found by sampling the frame function directly and printing scale values, not by re-reading code. (b) Import side effects: importing render-plan to reuse beatToGag executed its top-level main and re-rendered window chunks, corrupting a finished file. Fix: an invokedAsMain guard; corrupt partials deleted after ffprobe verification. (c) Swallowed exit codes: piping the audit through tail made the shell report tail's success, hiding

20 failed beats ('motion-exit:0' lie). Fix: redirect to a log file and read \$? / PIPESTATUS. Lesson repeated in all three: verify the artifact, never the assumption.

3.8 Slow renders and contention

Measured rate is about 1 frame per second (SVG raster per frame), so a 30-second window costs 12-14 minutes and the full 10 minutes costs hours. Consequences enforced as rules: renders run strictly sequentially, never two encoders at once; never start a duplicate of a running render (a second loop only steals RAM and slows both); the full stitch runs only after the window under judgment passes. Free memory is checked before heavy runs on this 2 GB-class device.

4. The verification loop (run in order, every change)

- 1. tsc --noEmit: types must be clean before anything runs.
- 2. compile-plan: SRT in, ShowPlan.json out. Overrides (unstageable beats) must stay low; each is an escape-hatch candidate, counted not hidden.
- 3. critic-plan: staging gates. Coverage, collision, timing/tiling, empty-tableau style, shot budget. Required result: ALL GATES PASS.
- 4. audit-motion: the anti-statue gate. Every beat's frame function is sampled at 8 timestamps and scored on 7 motion channels: figure travel (walk), rotation (sway/gesture), scale (pop/breath), talking (mouth), pointing (punch gesture), foreground change (prop action), camera (drift/punch). Each beat needs at least 4. Required result: ALL BEATS MOVE. This is the gate that would have rejected the W04 statue.
- 5. Render stills first (--stills at 3-4 timestamps): judge frames in seconds instead of re-rendering minutes.
- 6. Render the window, then prove movement on the pixels: decode at 2 fps to small gray frames and take the mean absolute pixel difference of every consecutive pair. Any frozen half-second (diff under 1.0) fails. Report mean/min over the window.
- 7. ffmpeg: streams must be H.264 video plus AAC audio, duration must match the window.
- 8. Watch it. Numbers catch statues and dead air; only eyes judge comedy timing. The 30-second test is judged before any full-duration render.

5. Recipe: hand-build a 30-second window

- Step 1: slice. Take the SRT cues for the span (about 30 s) and the matching voiceover slice. Note each cue's start/end: these are your talk windows.
- Step 2: beat it. Split at cue ends into 4-5 beats of 6-12 s. Name each beat's joke: setup or payoff, who acts, what changes. One change per beat minimum.
- Step 3: cast and props. Host at mark 400. Guests on fixed marks by side (left: 680; right: 990/1300/1620), entering from the nearest edge. Pick at most 2 props from the registry per beat; if the joke needs a new object, write a small builder function (pure SVG of t) rather than forcing a wrong one.
- Step 4: write each fn(t, lt). Skeleton per beat: host talk window covering the cues; one walk-in; prop cues at fixed lt times with easeIO windows; expression and gesture flip exactly at punchAt; camera wide with slow push, cut to tight on the punch.
- Step 5: stills, then render, then gates 6-8 above. If a beat fails the motion audit, the fix is staging (add business), never thresholds (never lower the bar to pass).

```
function wXa(t: number, lt: number): GagOut {
  const figs: Fig[] = [
    { id: 'host', st: { x: 400, y: 670, timeSec: t,
      isTalking: lt < 6.5, blink: blinkAt(t),
      expression: lt >= 6.5 ? 'smug_thumbs_up' : 'deadpan_classic',
      pointTarget: lt >= 6.5 ? { x: 1350, y: 480 } : undefined } },
    // ... guests with walkerX/walkerXR entrances ...
  ];
  let fg = propCues(t, lt); // strobes, drops, whips: f(lt)
  const punch = lt >= 6.5;
  return { bg, figs, fg,
    cam: punch ? { cx: 1150, cy: 560, zoom: 1.5, cut: false }
      : { cx: 960, cy: 560, zoom: 1.25, cut: lt < 0.05 } };
}
```

```
}
```

6. Hand-driven vs plan-driven

The plan pipeline (compile-plan -> critic -> audit-motion -> render-plan) exists so a finished script plus voiceover stages itself: it cuts windows, classifies beats, resolves props, stages cast, frames cameras. The hand-driven approach is the gold standard it is measured against and the escape hatch it falls back to: any beat the plan cannot stage is flagged as an override and can be hand-authored as a GagDef with identical settings (720p24, fixed GOP) so it concatenates cleanly. Rule of thumb: plan stages everything, hands fix heroes. A window is done when the audit passes, the frame-diff proves motion, and the 30-second test is judged good. Only then does it join the full stitch.